

### SAFETY DATA SHEET

Revision Date 15-Feb-2016

Revision Number 1

#### 1. Identification

**Product Name** TB Quad II

**Cat No. :** R03443

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**

**Company**

Remel  
12076 Santa Fe Drive  
Lenexa, KS 66215 United States  
Telephone: 1-800-255-6730  
Fax:1-800-621-8251

**Emergency Telephone Number**

INFOTRAC - 24 Hour Number: 1-800-535-5053  
Outside of the United States, call 24 Hour Number: 001-352-323-3500 (Call Collect)

#### 2. Hazard(s) identification

**Classification**

Classification under 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Based on available data, the classification criteria are not met

**Label Elements**

None required

**Hazards not otherwise classified (HNOC)**

None identified

#### 3. Composition / information on ingredients

| Component                       | CAS-No     | Weight % |
|---------------------------------|------------|----------|
| Glycerin                        | 56-81-5    | 0.49     |
| Oleic acid                      | 112-80-1   | Trace    |
| C.I. Acid blue 9, disodium salt | 3844-45-9  | Trace    |
| Rifampicin                      | 13292-46-1 | Trace    |

|                |         |       |
|----------------|---------|-------|
| Methyl alcohol | 67-56-1 | Trace |
|----------------|---------|-------|

#### 4. First-aid measures

|                                                               |                                                                                                                         |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Obtain medical attention.      |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                                             | Move to fresh air. Get medical attention immediately if symptoms occur.                                                 |
| <b>Ingestion</b>                                              | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | None reasonably foreseeable.<br>Treat symptomatically                                                                   |

#### 5. Fire-fighting measures

|                                         |                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                 |
| <b>Flash Point</b>                      | No information available °C / °F                                         |
| <b>Method -</b>                         | No information available                                                 |
| <b>Autoignition Temperature</b>         | No information available                                                 |
| <b>Explosion Limits</b>                 |                                                                          |
| <b>Upper</b>                            | No data available                                                        |
| <b>Lower</b>                            | No data available                                                        |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                 |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                 |

#### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

None known

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 1             | 0                   | 0                  | N/A                     |

#### 6. Accidental release measures

|                                             |                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment. Avoid dust formation.                        |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional ecological information.           |
| <b>Methods for Containment and Clean Up</b> | Sweep up or vacuum up spillage and collect in suitable container for disposal. Provide adequate ventilation. |

#### 7. Handling and storage

|                 |                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment. Ensure adequate ventilation. Avoid contact with skin, eyes and clothing. Avoid ingestion and inhalation. Avoid dust formation. |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component      | ACGIH TLV                             | OSHA PEL                                                                                                                                                                                 | NIOSH IDLH                                                                                                   |
|----------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Glycerin       |                                       | (Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>                                                       |                                                                                                              |
| Methyl alcohol | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin | (Vacated) TWA: 200 ppm<br>(Vacated) TWA: 260 mg/m <sup>3</sup><br>(Vacated) STEL: 250 ppm<br>(Vacated) STEL: 325 mg/m <sup>3</sup><br>Skin<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | IDLH: 6000 ppm<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 325 mg/m <sup>3</sup> |

| Component      | Quebec                                                                                             | Mexico OEL (TWA)                                                                           | Ontario TWA/EL                        |
|----------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------|
| Glycerin       | TWA: 10 mg/m <sup>3</sup>                                                                          | TWA: 10 mg/m <sup>3</sup>                                                                  |                                       |
| Methyl alcohol | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>Skin | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 310 mg/m <sup>3</sup> | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: The National Institute for Occupational Safety and Health Immediately Dangerous to Life or Health

**Engineering Measures**

None under normal use conditions.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                                  |
|----------------------------------|----------------------------------|
| Physical State                   | Solid Gel Consistency            |
| Appearance                       | No information available         |
| Odor                             | No information available         |
| Odor Threshold                   | No information available         |
| pH                               | No information available         |
| Melting Point/Range              | No data available                |
| Boiling Point/Range              | No information available         |
| Flash Point                      | No information available °C / °F |
| Evaporation Rate                 | No information available         |
| Flammability (solid,gas)         | No information available         |
| Flammability or explosive limits |                                  |
| Upper                            | No data available                |
| Lower                            | No data available                |
| Vapor Pressure                   | No information available         |

|                                        |                          |
|----------------------------------------|--------------------------|
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| VOC Content(%)                         | 0.54                     |

## 10. Stability and reactivity

|                                         |                                                           |
|-----------------------------------------|-----------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                |
| <b>Stability</b>                        | Stable under normal conditions.                           |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Avoid dust formation. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                   |
| <b>Hazardous Decomposition Products</b> | None under normal use conditions                          |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                  |
| <b>Hazardous Reactions</b>              | None under normal processing.                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|                    |                                                                               |
|--------------------|-------------------------------------------------------------------------------|
| <b>Oral LD50</b>   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Dermal LD50</b> | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Mist LC50</b>   | Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.     |
| <b>Vapor LC50</b>  | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |

#### Component Information

| Component      | LD50 Oral                                              | LD50 Dermal                                         | LC50 Inhalation                                                                      |
|----------------|--------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------|
| Glycerin       | LD50 = 12600 mg/kg ( Rat )                             | LD50 > 10 g/kg ( Rabbit )                           | LC50 > 570 mg/m <sup>3</sup> ( Rat ) 1 h                                             |
| Oleic acid     | LD50 = 25 g/kg ( Rat )                                 | Not listed                                          | Not listed                                                                           |
| Rifampicin     | LD50 = 1570 mg/kg ( Rat )                              | Not listed                                          | Not listed                                                                           |
| Methyl alcohol | Calc. ATE 60 mg/kg<br>LD50 > 1187 – 2769 mg/kg ( Rat ) | Calc. ATE 60 mg/kg<br>LD50 = 17100 mg/kg ( Rabbit ) | Calc. ATE 0.6 mg/L (vapours) or<br>0.5 mg/L (mists)<br>LC50 = 128.2 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                       | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------|------------|------------|------------|------------|------------|------------|
| Glycerin                        | 56-81-5    | Not listed | Not listed | Not listed | Not listed | Not listed |
| Oleic acid                      | 112-80-1   | Not listed | Not listed | Not listed | Not listed | Not listed |
| C.I. Acid blue 9, disodium salt | 3844-45-9  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Rifampicin                      | 13292-46-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                                   |                                                                |            |            |            |            |            |
|---------------------------------------------------|----------------------------------------------------------------|------------|------------|------------|------------|------------|
| Methyl alcohol                                    | 67-56-1                                                        | Not listed | Not listed | Not listed | Not listed | Not listed |
| <b>Mutagenic Effects</b>                          | No information available                                       |            |            |            |            |            |
| <b>Reproductive Effects</b>                       | No information available.                                      |            |            |            |            |            |
| <b>Developmental Effects</b>                      | No information available.                                      |            |            |            |            |            |
| <b>Teratogenicity</b>                             | No information available.                                      |            |            |            |            |            |
| <b>STOT - single exposure</b>                     | None known                                                     |            |            |            |            |            |
| <b>STOT - repeated exposure</b>                   | None known                                                     |            |            |            |            |            |
| <b>Aspiration hazard</b>                          | No information available                                       |            |            |            |            |            |
| <b>Symptoms / effects, both acute and delayed</b> | No information available                                       |            |            |            |            |            |
| <b>Endocrine Disruptor Information</b>            | No information available                                       |            |            |            |            |            |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. |            |            |            |            |            |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component      | Freshwater Algae | Freshwater Fish                                      | Microtox                                                                        | Water Flea                            |
|----------------|------------------|------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------|
| Glycerin       | Not listed       | LC50: 51 - 57 mL/L, 96h static (Oncorhynchus mykiss) | Not listed                                                                      | EC50: > 500 mg/L, 24h (Daphnia magna) |
| Oleic acid     | Not listed       | LC50: = 205 mg/L, 96h static (Pimephales promelas)   | Not listed                                                                      | Not listed                            |
| Methyl alcohol | Not listed       | Pimephales promelas: LC50 > 10000 mg/L 96h           | EC50 = 39000 mg/L 25 min<br>EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min | EC50 > 10000 mg/L 24h                 |

**Persistence and Degradability** No information available  
**Bioaccumulation/ Accumulation** No information available.

### Mobility

| Component      | log Pow |
|----------------|---------|
| Glycerin       | -1.76   |
| Oleic acid     | 7.73    |
| Methyl alcohol | -0.74   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                | RCRA - U Series Wastes | RCRA - P Series Wastes |
|--------------------------|------------------------|------------------------|
| Methyl alcohol - 67-56-1 | U154                   | -                      |

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

All of the components in the product are on the following Inventory lists: X = listed

#### International Inventories

| Component                       | TSCA | DSL | NDSL | EINECS    | ELINCS | NLP | PICCS | ENCS | AICS | IECSC | KECL |
|---------------------------------|------|-----|------|-----------|--------|-----|-------|------|------|-------|------|
| Glycerin                        | X    | X   | -    | 200-289-5 | -      |     | X     | X    | X    | X     | X    |
| Oleic acid                      | X    | X   | -    | 204-007-1 | -      |     | X     | X    | X    | X     | X    |
| C.I. Acid blue 9, disodium salt | X    | X   | -    | 223-339-8 | -      |     | X     | X    | X    | X     | X    |
| Rifampicin                      | -    | X   | -    | 236-312-0 | -      |     | X     | X    | X    | X     | X    |
| Methyl alcohol                  | X    | X   | -    | 200-659-6 | -      |     | X     | X    | X    | X     | X    |

Legend:

X - Listed

E - Indicates a substance that is the subject of a Section 5(e) Consent order under TSCA.

F - Indicates a substance that is the subject of a Section 5(f) Rule under TSCA.

N - Indicates a polymeric substance containing no free-radical initiator in its inventory name but is considered to cover the designated polymer made with any free-radical initiator regardless of the amount used.

P - Indicates a commenced PMN substance

R - Indicates a substance that is the subject of a Section 6 risk management rule under TSCA.

S - Indicates a substance that is identified in a proposed or final Significant New Use Rule

T - Indicates a substance that is the subject of a Section 4 test rule under TSCA.

XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)).

Y1 - Indicates an exempt polymer that has a number-average molecular weight of 1,000 or greater.

Y2 - Indicates an exempt polymer that is a polyester and is made only from reactants included in a specified list of low concern reactants that comprises one of the eligibility criteria for the exemption rule.

#### U.S. Federal Regulations

TSCA 12(b) Not applicable

#### SARA 313

| Component      | CAS-No  | Weight % | SARA 313 - Threshold Values % |
|----------------|---------|----------|-------------------------------|
| Methyl alcohol | 67-56-1 | Trace    | 1.0                           |

#### SARA 311/312 Hazard Categories

|                                   |    |
|-----------------------------------|----|
| Acute Health Hazard               | No |
| Chronic Health Hazard             | No |
| Fire Hazard                       | No |
| Sudden Release of Pressure Hazard | No |
| Reactive Hazard                   | No |

CWA (Clean Water Act) Not applicable

#### Clean Air Act

| Component      | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|----------------|-----------|-------------------------|-------------------------|
| Methyl alcohol | X         |                         | -                       |

OSHA Occupational Safety and Health Administration

Not applicable

#### CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

| Component      | Hazardous Substances RQs | CERCLA EHS RQs |
|----------------|--------------------------|----------------|
| Methyl alcohol | 5000 lb                  | -              |

California Proposition 65 This product contains the following proposition 65 chemicals

| Component  | CAS-No     | California Prop. 65 | Prop 65 NSRL | Category      |
|------------|------------|---------------------|--------------|---------------|
| Rifampicin | 13292-46-1 | Developmental       | -            | Developmental |

|                |         |                     |   |               |
|----------------|---------|---------------------|---|---------------|
|                |         | Female Reproductive |   |               |
| Methyl alcohol | 67-56-1 | Developmental       | - | Developmental |

#### U.S. State Right-to-Know Regulations

| Component                       | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------------------------|---------------|------------|--------------|----------|--------------|
| Glycerin                        | X             | X          | X            | -        | X            |
| Oleic acid                      | -             | -          | X            | -        | X            |
| C.I. Acid blue 9, disodium salt | X             | -          | -            | -        | -            |
| Methyl alcohol                  | X             | X          | X            | X        | X            |

#### U.S. Department of Transportation

Reportable Quantity (RQ): N  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

#### U.S. Department of Homeland Security

This product does not contain any DHS chemicals.

#### Other International Regulations

Mexico - Grade No information available

#### Canada

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR

WHMIS Hazard Class Non-controlled

### 16. Other information

Prepared By Regulatory Affairs  
 Thermo Fisher Scientific  
 Email: EMSDS.RA@thermofisher.com

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Revision Summary This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS)

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**